

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

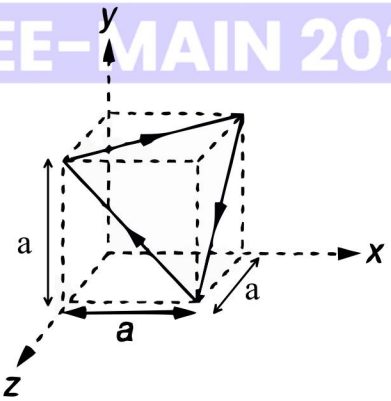
Sec: SR.IIT_*CO-SC(MODEL-A& APEX)		**Date:29-12-24**
Time: 3HRS	**GTM-5**	**Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The magnetic moment associated with the loop carrying current I_0 as shown in figure is



A) $\frac{3\sqrt{3}}{2}I_0a^2$ B) $\frac{2\sqrt{3}}{2}I_0a^2$ C) $\frac{2}{3}I_0a^2$ D) $\frac{\sqrt{3}}{2}I_0a^2$

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2. What mass (approximately) of coal with calorific value of 30 kJ/g is thermally equivalent to the heat liberated during the formation of one gram of He^4 from deuterium H^2?

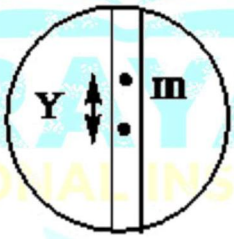
$m(H^2) = 2.01410$ amu, $m(He^4) = 4.002603$ amu

A) $1\times10^3\,kg$ B) $2\times10^4\,kg$ C) $3\times10^3\,kg$ D) $4\times10^4\,kg$

3. A lamp emits monochromatic green light uniformly in all directions. The lamp is 3% efficient in converting electrical power to electromagnetic waves and consumes 100 W of power. The amplitude of the electric field associated with the electromagnetic radiation at a distance of 10 m from the lamp will be

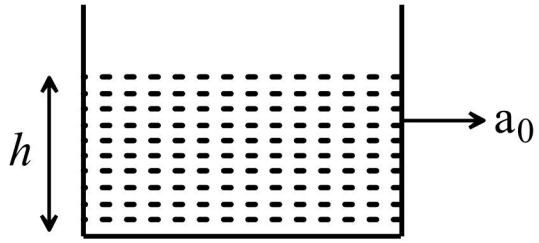
A) 1.34 V/m B) 2.68 V/m C) 5.36 V/m D) 9.37 V/m

4. Suppose a narrow tunnel is dug along the diameter of the earth assumed to be a sphere of uniform mass having density ρ. If a body of mass m is thrown in this tunnel, its acceleration at a distance y from the centre is given by



A) $\frac{4\pi}{3}g\rho y$ B) $\frac{3\pi}{4}G\rho y$ C) $\frac{4\pi}{3}\rho y$ D) $\frac{4\pi}{3}G\rho y$

5. A vessel contains liquid of density ρ up to height h. Vessel is given horizontal acceleration a_0 such that free surface of liquid makes an angle of 30° with horizontal. Value of a_0 is



A) g/2 B) g/3 C) $\frac{g}{\sqrt{3}}$ D) $\frac{\sqrt{3}g}{2}$

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6. If two soap bubbles of different radii are connected with each other through a pipe, then

A) air flows from the larger bubble into the smaller one until their sizes become equal.

B) the sizes of the bubbles remain unchanged.

C) air flows from the smaller bubbles into the larger one and the larger bubble grows at the expense of the smaller one.

D) air flows from the larger into the smaller bubble until their radii interchange.

7. A copper wire and a steel wire of the same diameter and length are connected end to end along the same line. A force is applied which stretches their combined length by 1 cm and the composite wire is at rest. Then the two wires have

A) the same stress and strain

B) the same stress but different strains

C) the same strain but different stresses

D) different stresses and strains

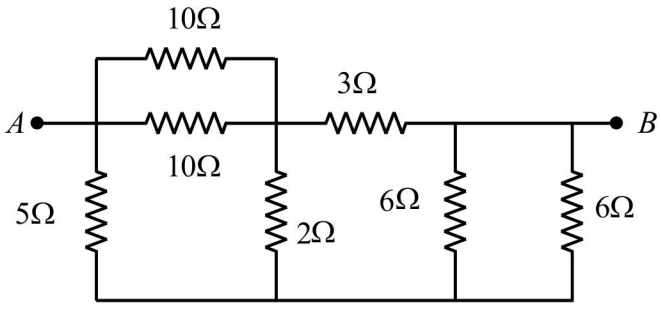
8. An energy of 24.6eV is required to remove one of the electrons from a helium atom which is in ground state. The energy (in eV) required to remove both the electrons from a neutral helium atom which is in ground state is

A) 38.2 B) 49.2 C) 51.8 D) 79.0

9. If 200 MeV energy is released per fission of $_{92}U^{235}$, how many fissions must occur per second to produce a power of 1 mW?

A) 6.25×10^{7} B) 12.25×10^{7} C) 25×10^{7} D) 3.125×10^{7}

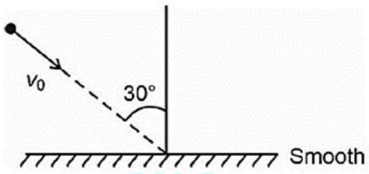
10. Find the equivalent resistance between points A and B (in Ω)



A) 4 B) 6 C) 11.2 D) 10.4

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11. A billiard ball of mass m moving with speed v_0 strikes a smooth floor at an angle of 30^0 with normal to floor. If ball rebounds at an angle of 60^0 with vertical, then coefficient of restitution is



A) 1/2 B) 1/3 C) $\frac{1}{2\sqrt{3}}$ D) $\frac{1}{3\sqrt{2}}$

12. A plano-convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered at the curved surface. Now this lens has been used to form the image of an object. At what distance from this lens an object is placed in order to have a real image of the size of the object?

A) 20cm B) 80cm C) 60cm D) 30cm

13. **Statement – 1:** Kinetic energy of a particle is conserved if it is acted upon by a conservative force only.

Statement–2: Work done by a conservative force in a closed path is zero.

(A) If both statements are true and the statement-2 is the correct explanation of statement – 1.

(B) If both statements are true but statement-2 is not the correct explanation of statement – 1.

(C) If statement – 1 is true but the statement – 2 is false

(D) If statement – 1 is false but the statement – 2 is true.

14. A thick-walled hollow sphere has outer radius R. It rolls down on an inclined plane without slipping and its speed at bottom is v_0. Now the incline is waxed so that the friction becomes zero. The sphere is observed to slide down without rolling and the speed at bottom now is $\frac{5v_0}{4}$. The radius of gyration of the hollow sphere about the axis through center is $\frac{nR}{4}$. Then the value of n is

A) 1 B) 4 C) 3 D) 2

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15. Electric potential V in volt in a region is given by $V = ax^2 + ay^2 + 2az^2$, where a is a constant. Work done by the field when a $2\mu c$ charge moves from point (0, 0, 0.1m) to origin is $5\times10^{-5} J$. Find a? (in V /m^2)

A) 1250 B) 1520 C) 1750 D) 1500

16. A projectile is projected with velocity 20 m/s at an angle of 53° with horizontal. Speed of the projectile when its velocity is perpendicular to its initial velocity, is $\left(g = 10\ m/s^2\right)$

A) 16 m/s B) 12 m/s C) 15 m/s D) 18 m/s

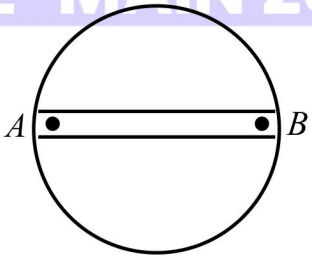
17. A wire of length $\ell = 6 \pm 0.06 cm$ and radius of cross-section $r = 0.5 \pm 0.005\ cm$ and mass $m = 0.3 \pm 0.003 gram$. Maximum percentage error in density is

A) 4 B) 2 C) 1 D) 6.8

18. In Young's double slit experiment using monochromatic light, the fringe patterns shifts by a certain distance on the screen when a mica sheet of refractive index 1.6 and thickness 1.964 micron is introduced in the path of one of the interfering waves. The mica sheet is then removed and the distance between the plane of slits and the screen is doubled. It is found that the distance between successive maxima now is the same as the observed fringe shift upon the introduction of the mica sheet. The wavelength of light is

A) 5762 Å B) 5825 Å C) 5892 Å D) 6500 Å

19. Suppose a narrow, and straight tunnel passes through the center of earth as shown in the figure. Two small balls A and B are released simultaneously at the two ends of the tunnel. The balls collide with each other and after collision one ball comes to rest and the other ball just manages to escape the earth's gravitational field. If the coefficient of restitution between the balls is e , then select the correct option.

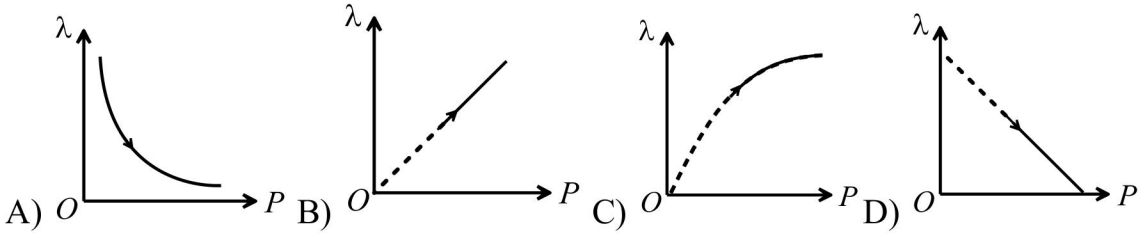


A) $e = \frac{\sqrt{3}}{2}$ B) $e = \frac{1}{\sqrt{2}}$

C) $\frac{1}{\sqrt{2}} < e < \frac{\sqrt{3}}{2}$ D) information is insufficient to predict

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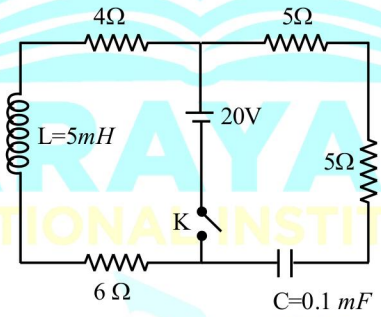
20. An ideal gas undergoes an isothermal process. The pressure (P) of the gas is plotted against the mean free path λ of the molecules. Select the correct graph.



SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

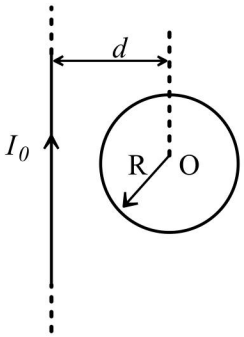
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. In the circuit shown, the key (K) is closed at $t = 0$, the current through the key after a long time is x (in Ampere). Then the value of x.



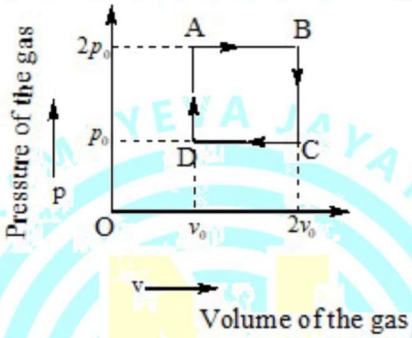
22. A uniform string of length 9 m and mass 4.5 kg is fixed at one end and hangs freely under its own weight. A wave pulse is generated at top end which runs down to other end. At the same moment a stone is released from rest and falls freely from the top of string. How far from the top does stone pass the wave (in m) ?

23. Current, I_0 flows in long straight conductor as shown. If magnetic field at center of circular loop in the same plane is zero, then current in the circular loop is $\frac{|n| I_0 R}{\pi d}$, then the value of |n| is

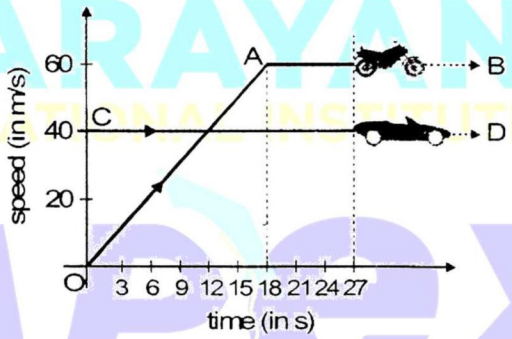


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24. The p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monoatomic gas. The amount of heat extracted from the source in a single cycle is $\left(\frac{x}{2}\right)p_0v_0$. Find the value of x.



25. At the instant a motor bike starts from rest in a given direction, a car overtakes the motor bike, both moving in the same direction. The speed time graphs for motor bike and car are represented by OAB and CD respectively. Then at t = 18s find the distance between then motor bike and car (in m).



CHEMISTRY **MAX.MARKS: 100**

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Which of the following is arranged in order of increasing radius?

A) $K^+(aq) < Na^+(aq) < Li^+(aq)$ B) $Na^+(aq) < K^+(aq) < Li^+(aq)$

C) $K^+(aq) < Li^+(aq) < Na^+(aq)$ D) $Li^+(aq) < Na^+(aq) < K^+(aq)$

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27. The correct increasing order of Cl – O bond order is :

A) $ClO_3^- < ClO_4^- < ClO_2^- < ClO^-$ B) $ClO^- < ClO_4^- < ClO_3^- < ClO_2^-$

C) $ClO^- < ClO_2^- < ClO_3^- < ClO_4^-$ D) $ClO_4^- < ClO_3^- < ClO_2^- < ClO^-$

28. Which species has maximum number of $(d_\pi - p_\pi)$ bonds?

A) I_3^- B) PO_4^{3-} C) SO_4^{2-} D) ClO_2^+

29. The solubility of $N_2(g)$ in water exposed to the atmosphere, when the partial pressure of N_2 is 0.75 atm, is $5.3\times10^{-4}M$. Its solubility at 1.0 atm and at the same temperature is

A) $7.07\times10^{-4}M$ B) $3.98\times10^{-4}M$ C) $5.10\times10^{-4}M$ D) $1.50\times10^{-4}M$

30. Which of the following statement is **INCORRECT**?

A) Thin film of polyacetylene can be used as electrodes in batteries.

B) Novocain a synthetic amine is used as an anaesthetic in dentistry

C) For measuring the resistance of ionic solutions alternating current (AC) source of power is used.

D) In the reaction at cathode of Leclanche cell, manganese is reduced from +4 oxidation state to +2 state.

31. Correct increasing order for the wavelengths of absorption in the visible region for the complexes of Co^{3+} is :

A) $\left[Co(H_2O)_6\right]^{3+}, \left[Co(en)_3\right]^{3+}, \left[Co(NH_3)_6\right]^{3+}$

B) $\left[Co(H_2O)_6\right]^{3+}, \left[Co(NH_3)_6\right]^{3+}, \left[Co(en)_3\right]^{3+}$

C) $\left[Co(NH_3)_6\right]^{3+}, \left[Co(en)_3\right]^{3+}, \left[Co(H_2O)_6\right]^{3+}$

D) $\left[Co(en)_3\right]^{3+}, \left[Co(NH_3)_6\right]^{3+}, \left[Co(H_2O)_6\right]^{3+}$

32. In which of the following pairs both the complexes show optical isomerism?

A) $\left[Ni(CO)_2(NO)_2\right]$, $cis-\left[Co(NH_3)_4Cl_2\right]$

B) $\left[Co(en)_3\right]Cl_3$, $cis-\left[Co(en)_2Cl_2\right]Cl$

C) $\left[PtCl(dien)\right]Cl, \left[NiCl_2Br_2\right]^{2-}$

D) $\left[Co(NO_3)_3(NH_3)_3\right]$, $cis-\left[Pt(en)_2Cl_2\right]$

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33. **Statement – 1** : The third period contains only 8 elements and not 18 like 4th period.

Statement – 2 : In third period filling starts from $3s^1$ and complete at $3p^6$, whereas in fourth period it starts from $4s^1$ and complete after $3d^{10}$ and $4s^2$.

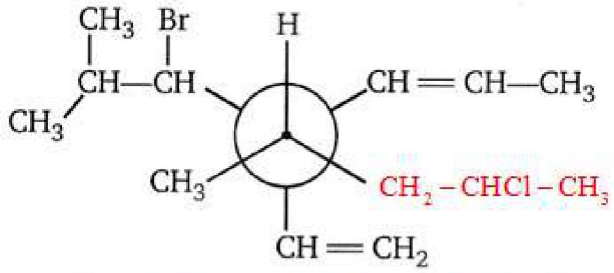
A) Both statements are true and statement-2 is the correct explanation of statement–1.

B) Both statements are true but statement-2 is not correct explanation of statement–1.

C) Statement – 1 is true but the statement–2 is false

D) Statement – 1 is false but the statement–2 is true.

34. Total number of stereoisomers



A) 2^3 B) 2^4 C) 2^5 (D) 2^6

35. Among the following which is most reactive towards S_N2 reaction?

A) $CH_3 - Cl$ B) $H_2C = CH - Cl$ C) C_6H_5-Cl (D) $CH_3 - F$

36. Generally Alkyl chlorides in pure form are prepared by the action of $SOCl_2$ on alcohols because :

A) alkyl chlorides are immiscible with $SOCl_2$

B) In this reaction alkyl halide is formed along with gases SO_2 and HCl, the two gaseous products are escapable, hence the reaction gives pure alkyl halides

C) alcohol and $SOCl_2$ are soluble in water

D) the reaction does not occurs via intermediate formation of an alkyl chlorosulphite

37. Identify the set of reagents/reaction-conditions 'X' and 'Y' in the following set of transformation:

$$CH_3 - CH_2 - CH_2Br \xrightarrow{X} \text{Product} \xrightarrow{Y} CH_3 - \underset{\underset{Br}{|}}{CH} - CH_3$$

A) X = dilute aqueous NaOH, 20°C; Y = Br_2/CCl_4, 20°C

B) X = conc. alcoholic NaOH, 80°C; Y = HBr/CH_3COOH

C) X = dilute aqueous NaOH, 20°C; Y = $Br_2/CHCl_3$, 20°C

D) X = conc. alcoholic NaOH, 80°C; Y = $Br_2/CHCl_3$

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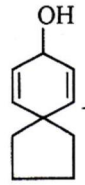
38. In which of the following processes involving ideal gas ΔS_{sys} is +ve (> 0)

A) Adiabatic irreversible expansion only

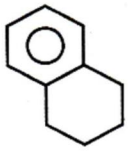
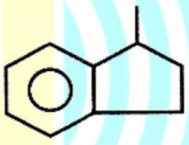
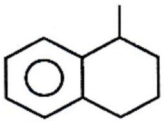
B) Adiabatic irreversible compression only

C) Adiabatic irreversible expansion and compression

D) Adiabatic reversible expansion and compression

39. 

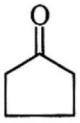
(A) ; Product (A) is:

(A)  (B)  (C)  (D) 

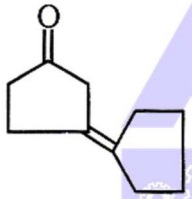
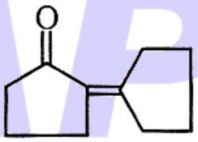
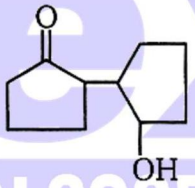
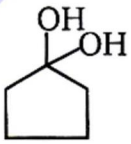
40. Which of the following is both acid-base and inter molecular redox reaction

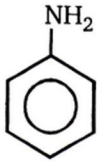
A) $Pb_3O_4 + HCl \rightarrow$ B) $Pb_3O_4 + HNO_3 \rightarrow$

C) $PbO + HCl \rightarrow$ D) $PbO_2 + HNO_3 \rightarrow$

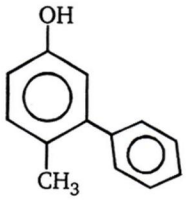
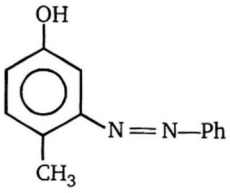
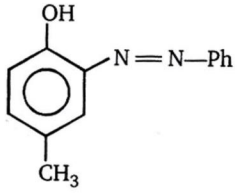
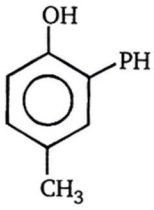
41. 

(A) ; Product (A) is:

(A)  (B)  (C)  (D) 

42. 

(B), Product (B) is:

A)  B)  C)  D) 

For More Material Join: @JEEAdvanced_2025

43. Which of the following pairs give positive Tollens' test ?

A) Glucose, sucrose B) Glucose, fructose

C) Hexanal, acetophenone D) Fructose, sucrose

44. The dissociation energies of H_2 and O_2 are 104 and 118 kcal mol^{-1} respectively. The heat of reaction $\frac{1}{2}H_2(g)+\frac{1}{2}O_2(g)\rightarrow O-H(g)$ is 10 kcal. The bond energy of O – H bond is:

A) 101 kcal/mol B) 111 kcal/mol C) 10.1 kcal/mol D) 11.1 kcal/mol

45. The change in entropy of 2 moles of an ideal gas upon isothermal expansion at 243.6 K from 20 L to the state where pressure becomes 1 atm is : (Given : ln 2 = 0.693, $R = 0.082\ lit\ atm\ mole^{-1}K^{-1}$, $R = 2\ cal\ \ mole^{-1}K^{-1}$)

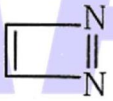
A) 1.385 cal/K B) – 1.2 cal/K C) 1.2 cal/K D) 2.77 cal/K

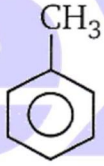
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

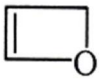
46. X = Number of compounds unstable at room temperature.

(a) 

(b) 

(c) 

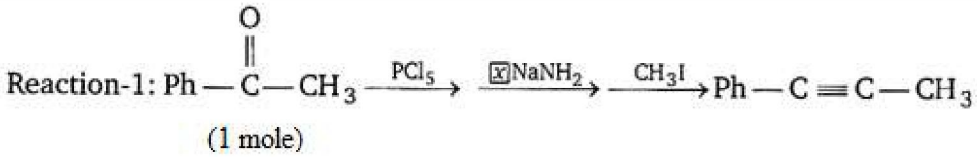
(d) 

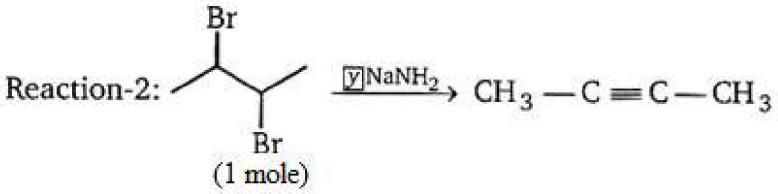
(e) 

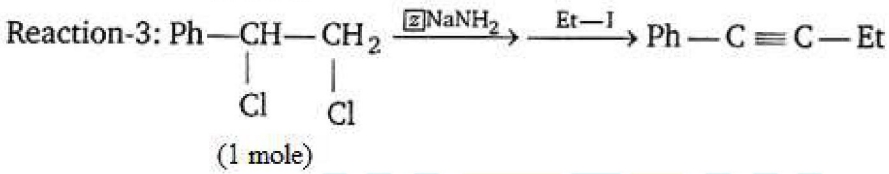
(f) 

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47.







x, y, z are moles used. Sum of [x + y + z =]

48. Calculate the pH of buffer solution prepared by dissolving 0.20 mole of sodium cyanate (NaCNO) and 1.0 mole of cynic acid (HCNO) in enough water to make 1.0 litre of solution, $K_a(HCNO) = 2.0 \times 10^{-4}$. (Given: $\log 2 = 0.3$; $\log 5 = 0.7$)

49. Osmotic pressure of aqueous urea solution at 293 K decreases from 400 mmHg to 105.3 mmHg due to dilution by 'x' times, and subsequent raising the temperature to 308 K. Thus, extent of dilution 'x' is

50. How many unpaired electrons will be in 4f subshell of Gadolinium ($Z = 64$)?

MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. The value of definite integral $\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x} dx$ is :

(A) $\frac{\pi}{2}$ (B) π (C) π^2 (D) $\frac{\pi^2}{2}$

52. If $\sum_{r=1}^{100} \sin^{-1}\left(\frac{1}{\sqrt{r^2+1}} \times \frac{1}{\sqrt{r^2+2r+2}}\right)$ is equal to $\tan^{-1}\left(\frac{p}{q}\right)$ where p and q are co-prime, then the value of (p + q) is equal to :

(A) 99 (B) 100 (C) 101 (D) 102

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53. If $y=\left(x+\sqrt{1+x^2}\right)^n$, then the value of $\left(1+x^2\right)\frac{d^2y}{dx^2}+x\frac{dy}{dx}$ is equal to :

(A) - y (B) $-n^2y$ (C) n^2y (D) $-ny^2$

54. If $f'(x)+(f'(x))^2+(f'(x))^3+(f'(x))^4+.....\infty=e^x$, where $f'(x)\in(-1,1)$ and f(0) = 0 then the value of $\lim\limits_{x\to0}(1+f(x))^{\frac{1}{x}}$ is equal to :

(A) e (B) $\sqrt{e}$ (C) $1/\sqrt{e}$ (D) 1

55. If $\int\frac{3\tan\left(x-\frac{\pi}{4}\right)}{\cos^2x\sqrt{\tan^3x+\tan^2x+\tan x}}dx=k\tan^{-1}\left(\sqrt{\tan x+1+\cot x}\right)+C$, then the value of k is [where C is constant of integration.]

(A) 2 (B) 3 (C) 6 (D) 8

56. The points (2, 5) and (5, 1) are two opposite vertices of a rectangle. If other two vertices are points on the straight line y = 2x + k, then the value of k is :

(A) 4 (B) 3 (C) - 4 (D) - 3

57. Let a function f(x) be defined in [- 2, 2] as $f(x)=\begin{cases}\{x\}, & -2\le x<-1\\ |\text{sgn}(x)|, & -1\le x\le1\\ \{-x\}, & 1<x\le2\end{cases}$, where {x} denotes fractional part, then area bounded by graph of f(x) and x-axis is :

(A) 2 (B) 3 (C) 4 (D) 5

58. For a constant k, the two roots of the quadratic equation $3x^2 - x + k = 0$ are $\sin\theta$ *and* $\cos\theta$. The value of $54\left(\sin^3\theta+\cos^3\theta\right)$, is :

(A) 25 (B) 26 (C) 27 (D) 28

59. Let $\vec{a},\vec{b},\vec{c}$ be three non-zero vectors satisfying $\vec{a}=\vec{b}\times\vec{c}+2\vec{b}$ where $\left|\vec{b}\right|=\left|\vec{c}\right|=2$ and $\left|\vec{a}\right|\le4$. The sum of possible value(s) of $\left|2\vec{a}+\vec{b}+\vec{c}\right|$ is :

(A) 8 (B) 12 (C) 20 (D) 32

60. The probability of occurrence of a multiple of 2 on one dice and a multiple of 3 on the other dice if both are thrown together, is :

(A) $\frac{7}{36}$ (B) $\frac{1}{3}$ (C) $\frac{1}{6}$ (D) $\frac{11}{36}$

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61. Let z be the complex number satisfying $|z + 16| = 4|z + 1|$, then :

(A) $|z| = 4$ (B) $|z| = 5$ (C) $|z| = 6$ (D) $4 < |z| < 64$

62. Let A, B are two square matrices of order 3×3 such that $ABA - 3B = 3A - AB^2$.

Consider following statements.

(S1) : If $A + B$ is non singular then A and B are non singular

(S2) : If $A + B$ is non singular then $\left(A^2 - B^2\right).\left(A + B\right)^{-1} = A - B$

A) (S1) is true but (S2) is false B) (S2) is true but (S1) is false

C) Both (S1) and (S2) are true D) Both (S1) and (S2) are false

63. There are 5 points P_1, P_2, P_3, P_4, P_5 on the side AB, excluding A and B, of a triangle ABC. Similarly there are 6 points P_6, P_7,…,P_{11} on the side BC and 7 points P_{12}, P_{13}, …, P_{18} on the side CA of the triangle. The number of triangles, that can be formed using the points P_1, P_2, …, P_{18} as vertices, is :

(A) 776 (B) 751 (C) 796 (D) 771

64. The equation of the lines joining the vertex of the parabola $y^2 = 6x$ to the points on it which have abscissa 24 are :

(A) $y \pm 2x = 0$ (B) $3y \pm x = 0$ (C) $x \pm 2y = 0$ (D) $2x \pm y = 0$

65. Number of points of discontinuity of $f\left(x\right) = 12\left(\left\{\frac{x}{11}\right\} + 14\left[\frac{x}{13}\right]\right)\left(\left[x\right] - \left[\frac{x}{3}\right]\right)$ in $x \in \left(0, 25\right)$ is (Where $\left[x\right]$ denotes G.I.F and $\left\{x\right\}$ denotes fractional part of x)

(A) 16 (B) 15 (C) 14 (D) 13

66. Let $x = 111.....11$ (20 digits)

$y = 333.....33$ (10 digits)

and $z = 222.....22$ (10 digits) , then find the value of $\frac{x - y^2}{z}$

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{5}{9}$

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67. The centre of a circle passing through the points (0, 0), (1, 0) and touching the circle $x^2 + y^2 = 9$ can be

(A) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, \frac{3}{2}\right)$ (C) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, -\sqrt{2}\right)$

68. Let $A = \{x \in R : |x+1| < 2\}$ and $B = \{x \in R : |x-1| \geq 2\}$. Then which one of the following statements is NOT true ?

(A) A – B = (-1, 1) (B) B – A = R – (-3, 1)

(C) $A \cap B = (-3, -1]$ (D) $A \cup B = R - [1, 3)$

69. If the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the line $\frac{x}{7} + \frac{y}{2\sqrt{6}} = 1$ on the x-axis and the line $\frac{x}{7} - \frac{y}{2\sqrt{6}} = 1$ on the y-axis, then the eccentricity of the ellipse is

(A) $\frac{5}{7}$ (B) $\frac{2\sqrt{6}}{7}$ (C) $\frac{3}{7}$ (D) $\frac{2\sqrt{5}}{7}$

70. A wire of length 22 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into an equilateral triangle. Then, the length of the side of the equilateral triangle, so that the combined area of the square and the equilateral triangle is minimum, is :

(A) $\frac{22}{9 + 4\sqrt{3}}$ (B) $\frac{66}{9 + 4\sqrt{3}}$ (C) $\frac{22}{4 + 9\sqrt{3}}$ (D) $\frac{66}{4 + 9\sqrt{3}}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. Let S be the set of all values of λ, for which the shortest distance between the lines $\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ and $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ is 13. Then $8\left|\sum_{\lambda \in S} \lambda\right|$ is equals to

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72. Let $f : R - \{1\} \to R - \{1\}$ be a function satisfying the differential equation $2x(y + x)dx - x^2(dx + dy) = (x + y)^2dx$ with f(2) = 2. If area enclosed by y = f(x) and x-axis from x = 2 to x = 3 is (a + ln b) where $a, b \in N$, then find the value of (a + b).

73. If term independent of x in the expansion of $\left(x^2 - \frac{3}{x}\right)^6$ *is* $\lambda 3^5$, then find the value of λ .

74. Let $f(x) = \frac{1}{\cos^2 x} + \frac{4}{\sin^2 x}$. The minimum value of $f(x)$ *for* $0 < x < \frac{\pi}{2}$ is :

75. The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5. Then, the correct variance is equal to ____

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